

The Anchor Protocol:

A Formally Verified Control Plane for Local Agent Swarms

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Abstract

As AI agents transition from isolated copilots to collaborative, autonomous swarms, local development environments face a crisis of coordination and trust. Existing tools designed for human-driven, static infrastructures lack the primitives required to prevent dynamic agents from corrupting shared state or squatting on ephemeral ports.

This paper introduces the **Anchor Protocol**, a cryptographic and semantic identity framework built into the Port Daddy daemon. We detail the protocol’s evolution from symmetric MACs to asymmetric Ed25519 signatures and multi-hop delegation chains (inspired by Macaroons [4]). Furthermore, we present the formal verification of the protocol’s design using the **ProVerif** symbolic analyzer [1] and the mathematical proof of its implementation, demonstrating immunity to algorithm confusion [10], impersonation, and timing side-channels. We are precise about scope: the multi-hop delegation soundness is machine-checked at the chain depth realized in deployment (depth 3), and we state explicitly (§7.3) that extending the symbolic proof to unbounded chain depth—as distinct from the unbounded *sessions* ProVerif already quantifies over—remains future work.

Keywords: distributed systems security, formal verification, capability-based security, multi-agent coordination, symbolic protocol analysis, ProVerif, JWT security, Ed25519

Reading time: approximately 30 minutes end-to-end. The companion paper, The Bonded Commons: Pre-Transactional Trust Infrastructure for Multi-Agent Systems (Owens 2026), carries the economic and governance argument; either paper can be read first.

THE PORT DADDY TRILOGY

PAPER II OF III

I. The Bonded Commons. *Why* mutually-suspicious agents need a pre-transactional *commons authority*: bonded participation, an immutable evidence trail, and a value-conservation law. The economic and governance case for coordination at all.

II. The Anchor Protocol ([this paper](#)). *How* a single process proves its identity and attenuates its capability to the local daemon — a formally verified (ProVerif, Kani) capability substrate, scoped to one machine.

III. The Federated Harbor. How identity, revocation, and settlement compose *across sovereign machines* when no operator is a root of authority for any other — the federation layer.

You are reading **Paper II**, the cryptographic substrate of the series; it can be read first or on its own.

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1 Introduction: The “Local Swarm” Problem

In a multi-agent development environment, agents operate concurrently to read files, spin up test servers, and execute commands. This introduces three critical threat vectors on `localhost`:

1. **Port Squatting (The “Ghost in the Harbor”)**: If Agent A crashes, a malicious or malfunctioning Agent B can bind to Agent A’s previously assigned port, intercepting traffic meant for A.
2. **Resource Contention**: Agents fighting over shared resources (e.g., a SQLite database file) without a centralized locking mechanism cause state corruption.
3. **Privilege Escalation**: A “Reviewer” agent delegated by a “Coder” agent should not possess the rights to initiate production deployments, yet local environments rarely enforce principle-of-least-privilege between processes.

To solve this, Port Daddy introduces the concept of a **Harbor**—a zero-trust, namespace-isolated execution environment where agents must prove their identity and capabilities. Our approach builds upon foundational work in capability-based security by Dennis and Van Horn [7] and the principle of least privilege articulated by Saltzer and Schroeder [15].

Companion paper. This paper is the cryptographic substrate. Its companion, *The Bonded Commons: Pre-Transactional Trust Infrastructure for Multi-Agent Systems*, builds the economic and game-theoretic layer on top: bonded participation, conditional-Pareto-dominant auctions, A5/A6 mechanism analysis, and the Sen-grounded justification for why coercive coordination is required at all. The Anchor Protocol is a self-contained subset; *The Bonded Commons* consumes it as the trust kernel for the wider mechanism design. Read this paper for “how does the daemon authenticate an agent?”; read the companion for “why should there be a daemon at all?”

2 The Anchor Protocol Architecture

The Anchor Protocol defines how agents authenticate to a Harbor and to each other. It issues **Harbor Cards** (highly constrained JSON Web Tokens) and evolves them across four phases, each adding a capability that closes a specific attack surface from the previous phase (Figure 1).

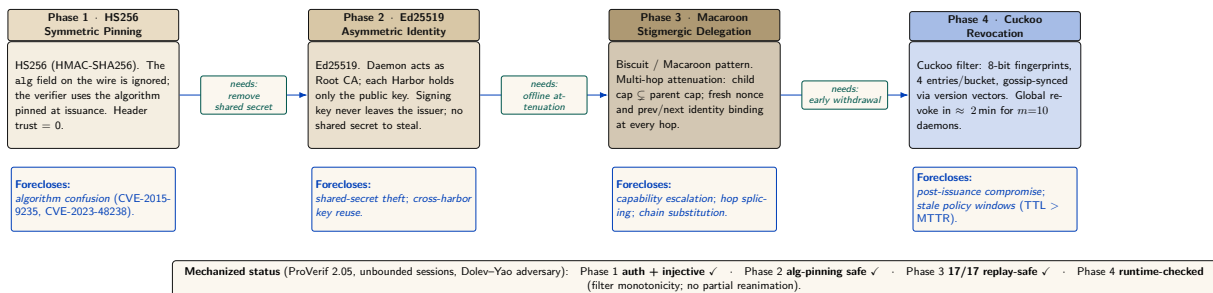


Figure 1: The four phases of the Anchor Protocol. Each phase is a refinement of the previous: the cryptographic primitive changes, the property strengthens, and a specific threat is foreclosed. The teal pill on each arrow records *why* the next phase was forced — not as historical narrative but as a constructive argument that no earlier phase suffices on its own. Phases 1–3 are mechanized in ProVerif (§A); Phase 4 is enforced at runtime against monotonicity and reanimation invariants (§2.4).

A common thread across all phases is *capability attenuation*: a delegated token never carries more authority than its parent, and TTL only shrinks. Figure 2 shows the nested-cap structure that the verifier enforces on every chain.

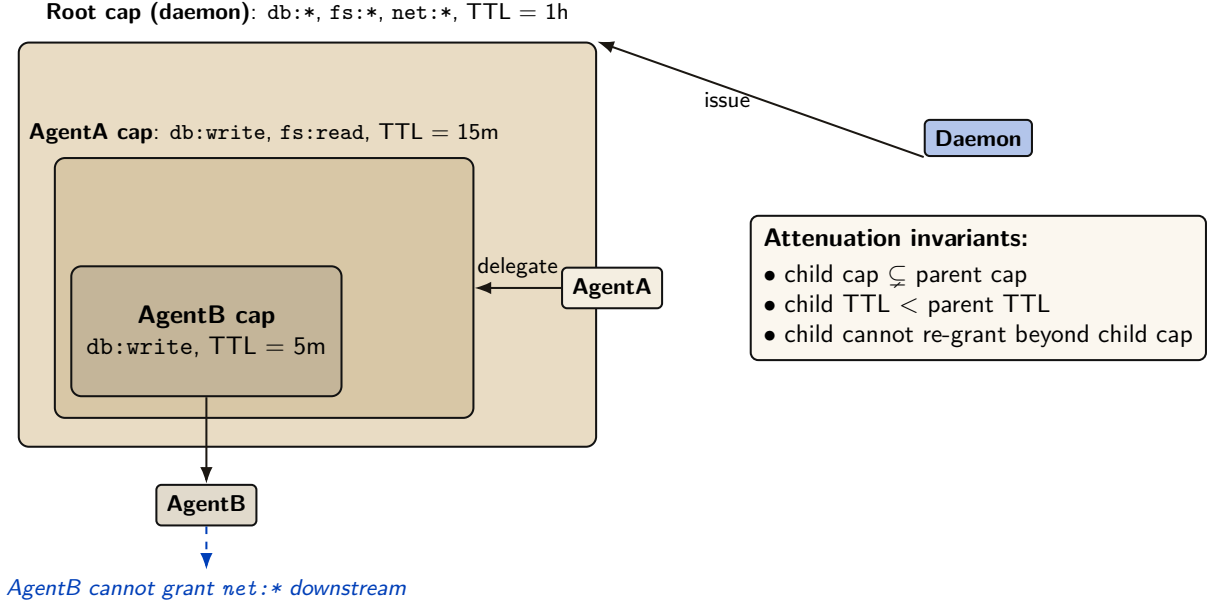


Figure 2: Capability attenuation under delegation. Each child capability is a strict subset of its parent’s (rights and TTL both shrink monotonically). AgentB inherits only what AgentA explicitly delegates, and cannot re-grant capabilities it does not hold. This is enforced both at issuance (parent signs over the child’s attenuated claim set) and at verification (the verifier checks the chain of `is_subset` relations).

2.1 Phase 1: Symmetric Pinning

Early versions of the protocol relied on HS256 (HMAC-SHA256). The primary vulnerability in JWTs is “Algorithm Confusion” (CVE-2015-9235 [11], CVE-2023-48238 [13]), where an attacker modifies the token header to `{"alg": "none"}` or symmetric HS256 while using an asymmetric public key as the HMAC secret.

Property 2.1 (Algorithmic Pinning). The Port Daddy Verifier employs strict **Algorithmic Pinning**. It entirely ignores the user-provided `alg` header and explicitly forces the underlying crypto library to evaluate the signature using the pre-determined algorithm. This approach is recommended by the JWT Best Current Practices draft [14].

Figure 3 shows the two verifier paths side by side: the naive verifier trusts the `alg` field on the wire (and is exploited), while the pinned verifier records the issuance algorithm out-of-band and admits no rewrite trace.

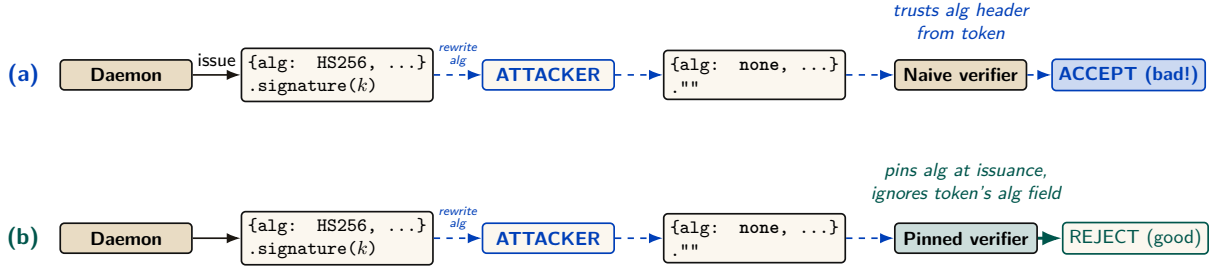


Figure 3: Algorithm-confusion attack and the pinned-verifier defense. (a) A naive verifier trusts the `alg` field of the incoming token; an attacker who can intercept the token rewrites `alg=HS256` to `alg=none`, strips the signature, and the verifier accepts an unauthenticated message. (b) The pinned verifier records the issuance algorithm out-of-band, then re-checks each token against the pinned algorithm rather than the token’s self-described one. The ProVerif model encodes both verifiers and shows the naive path produces a counter-trace witness for the rewrite, while the pinned path admits no such trace. Runtime conformance: 5 attack patterns + 2 controls, all pass.

2.2 Phase 2: Asymmetric Identity (Ed25519)

To support decentralized Agent-to-Agent (A2A) communication without sharing HMAC secrets, the protocol transitioned to **Ed25519 (EdDSA)** [8, 9]. Ed25519 provides fast, deterministic signatures with small keys and signatures, making it ideal for local agent communication.

Definition 2.1 (Harbor Key Hierarchy). The Port Daddy Daemon acts as the Root Certificate Authority (CA). Each Harbor is assigned a unique Ed25519 keypair. The Daemon issues Harbor Cards signed by the Harbor’s private key. Agents use the Harbor’s public key (broadcast via Port Daddy’s ambient discovery service) to verify tokens presented by peers.

2.3 Phase 3: Stigmergic Delegation (The “Biscuit” Pattern)

Port Daddy v4 introduces multi-hop delegation. When Agent A spawns Agent B, it does not need to request a new token from the Daemon. Instead, it uses **Offline Attenuation**, inspired by Macaroons [4].

Definition 2.2 (Offline Attenuation). Agent A takes its existing Harbor Card, appends a new set of restricted capabilities (e.g., `["db:read"]` reduced from `["db:read", "db:write"]`), and signs the *entire chain* with its own ephemeral private key.

The Harbor verifies the Daemon’s root signature, Agent A’s delegation signature, and mathematically ensures that Agent B’s capabilities are a strict subset of Agent A’s.

For depth- N chains, the verifier walks the structure shown in Figure 4. Each hop must produce a fresh nonce and bind the previous and next identities into its signature so that an attacker who intercepts a single hop cannot splice it into a different chain.

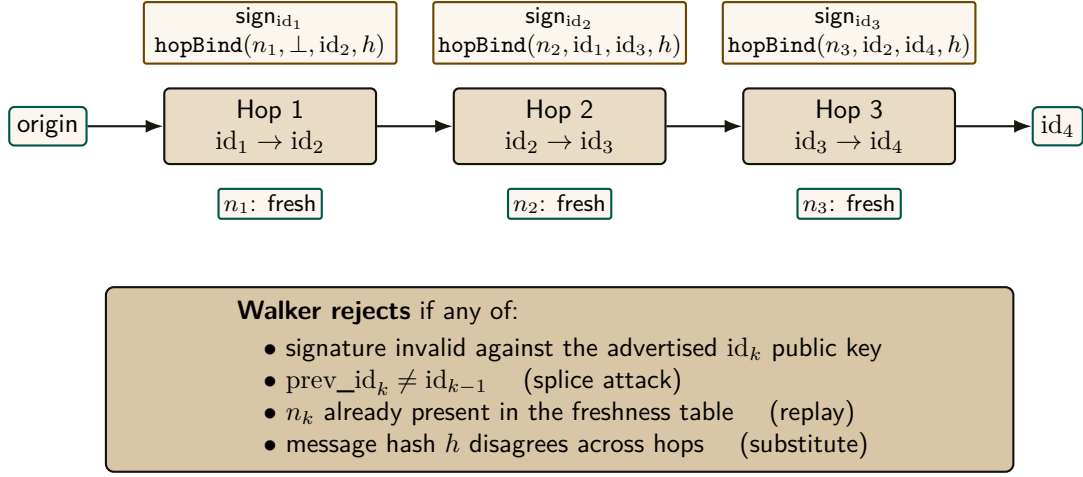


Figure 4: Multi-hop delegation walker. Each hop k signs the binding tuple (n_k, id_{k-1}, id_k, h) under Ed25519, where h is the message hash that must remain constant across the chain. The walker verifies depth- N chains with a nonce-freshness table and rejects splices, replays, hop-swaps, and substitution. Runtime: `lib/delegation-chain.ts`; ProVerif conformance test passes 17 / 17.

2.4 Phase 4: Revocation via Cuckoo Filter

Capability tokens have a TTL, but natural expiry is not always sufficient. Two circumstances force early withdrawal: (i) *compromise*—a Harbor Card leaks from its agent’s process, and every second until TTL is a second the attacker has legitimate capabilities; and (ii) *policy reversal*—a principal decides an agent should no longer hold `db:write`, and waiting for TTL is unacceptable. A naive revocation list is $O(n)$ per verification and grows monotonically; in a daemon mesh, propagating it is $O(n \cdot m)$ for m peers. The Anchor Protocol commits to a cuckoo filter [25] for revocation (Figure 5).

Cuckoo filters in one paragraph. A cuckoo filter is a compact probabilistic set: it answers “is x in this set?” with possible false positives but no false negatives, like a Bloom filter, but supports *deletions* and uses less space at the false-positive rates we care about. Each inserted key is reduced to a small fingerprint (8 bits here) and placed in one of two candidate buckets selected by two hashes; on collision, the resident fingerprint is “kicked” to its alternate bucket, recursively, until a slot opens or a kick budget is exhausted. Lookups check both candidate buckets for the fingerprint. The relevant properties for revocation are: $O(1)$ expected insert/lookup/delete, no false negatives (a revoked card is *never* missed), and a tunable false-positive rate that determines space per entry. Fan et al. prove the FP bound holds up to a load factor of ≈ 0.95 ; we enforce that ceiling at the insert path.

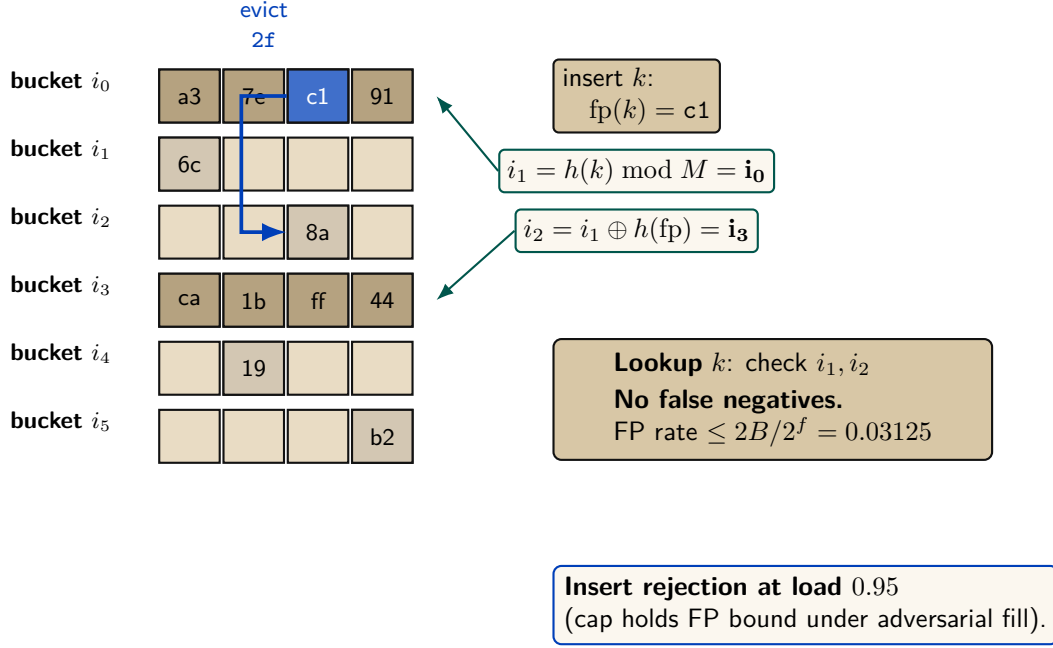


Figure 5: Cuckoo-filter revocation gate (Fan et al. [25]) with the load-factor ceiling that defeats the pollution attack. Insert lands at either of two candidate buckets $i_1, i_2 = i_1 \oplus h(fp)$; on full, evict and re-insert at the kicked fingerprint’s alternate. Insert rejection at 0.95 caps occupancy inside the regime where the false-positive bound holds.

Why cuckoo, not Bloom. For uniform-TTL workloads, a rotating Bloom filter would suffice. Port Daddy’s TTLs are heterogeneous—Shipwright proposals run hours to days, `pd spawn` children run minutes, one-shot agents run seconds. Rotating Bloom either rotates at the longest TTL (short-TTL revocations linger orders of magnitude longer than needed) or buckets by TTL class (reintroducing the structural complexity cuckoo collapses). Cuckoo provides $O(1)$ per-entry removal at each specific expiry, plus the same removal handles operator-initiated early-revocation undo cleanly.

Space. At false-positive rate ϵ , a cuckoo filter uses approximately $\log_2(1/\epsilon) + 3$ bits per entry. For $\epsilon = 10^{-3}$, ≈ 13 bits per revoked card. A fleet retaining 10^5 active revocations fits in ≈ 160 KB—trivially in memory, trivially gossiped.

Definition 2.3 (Revocation-Augmented Verification). Let R_t denote the set of revoked Harbor Card IDs at time t . Each daemon d maintains a cuckoo filter $F_d \approx R_t$. Every Harbor Card verification additionally:

1. Looks up the card’s `kid` in F_d . If absent, admit.
2. If present, query the local `revocations` table (authoritative). If genuinely revoked, reject with `revoked`; otherwise the filter is in the false-positive regime and the caller is admitted, with an asynchronous reconciliation job repopulating the filter.

Gossip-based synchronization. Daemons in a mesh synchronize filters by anti-entropy gossip [26]: each daemon maintains a version vector over peers, periodically picks a random peer, exchanges vectors, ships deltas (cards revoked since the peer’s last seen version), and applies them. A revocation reaches all m peers in expected $O(\log m)$ gossip rounds. For $m = 10$ daemons at 30-second intervals, all nodes see a new revocation within ≈ 2 minutes. Security-critical revocations can trigger immediate push (the issuer broadcasts to every known peer). Figure 6 traces a single revocation through a five-daemon mesh frame-by-frame.

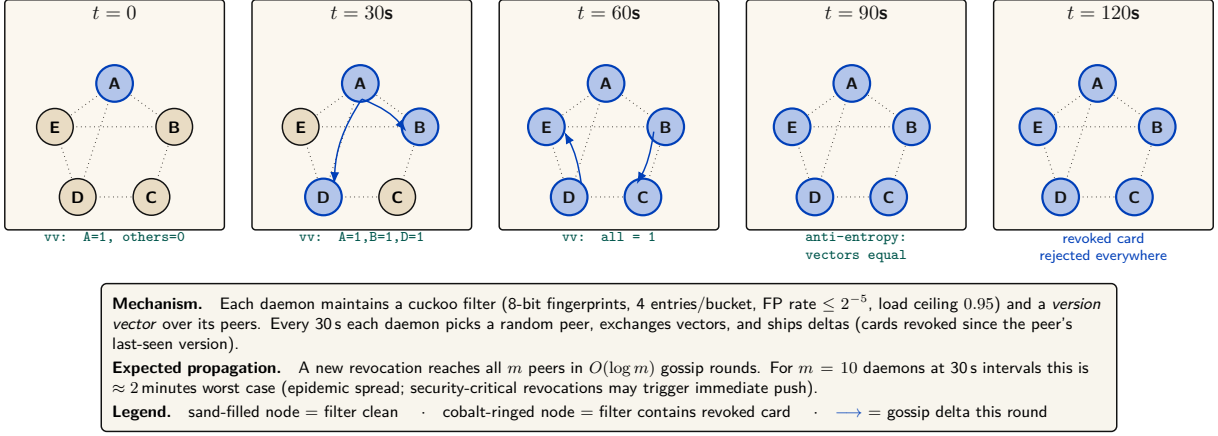


Figure 6: Gossip-synchronized revocation across a five-daemon mesh. At $t = 0$ daemon A locally revokes a Harbor Card (its filter and revocation table). Anti-entropy gossip ships deltas to two random peers every 30s, reaching all daemons in $O(\log m)$ rounds. For a ten-daemon mesh this gives global propagation in ≈ 2 minutes, well inside the MTTR window where leaked cards remain economically useful. Version vectors guarantee convergence even under reordered or duplicated delta deliveries.

Property 2.2 (Filter Monotonicity over a TTL Epoch). For any Harbor Card c with issue time t_0 and TTL τ , if $c \in R_t$ for some $t \in [t_0, t_0 + \tau]$, then $c \in R_{t'}$ for all $t' \in [t, t_0 + \tau]$. After $t_0 + \tau$, c may be removed from R since the card has expired.

Property 2.3 (No Partial Reanimation). A revoked card cannot be un-revoked within its TTL. If an operator decides the revocation was erroneous, they must issue a new card.

Property 2.4 (Filter Freshness Bound). Under the gossip protocol with period Δ in a mesh of m daemons, every daemon's filter converges to global R_t within expected time $\Delta \cdot (1 + \ln m)$.

Soundness preservation. Revocation strictly narrows acceptance, so existing ProVerif soundness proofs (Theorem 4.1) are unaffected. The new proof obligation is small: revocation is *enforced*, which follows by inspection of the verification path—the revocation check is unconditional and precedes admission. The card lifecycle gains a state (*valid* $\rightarrow$ *revoked* $\rightarrow$ *expired*) modeled as an additional transition.

Privacy. A Dolev-Yao adversary observing all gossip traffic learns the revoked-card IDs, leaking which agents have been disciplined. For an organization's own daemons, this is acceptable audit transparency. Stronger privacy is available by encoding revocations as salted hashes $H(\text{kid}, \text{salt})$ where the salt rotates daily, stored encrypted under a harbor session key.

3 Related Work

3.1 Formal Verification of Security Protocols

Formal verification of cryptographic protocols has matured significantly over the past two decades. Following the seminal work by Lowe [5] on authentication hierarchies and the Dolev-Yao model [6], several automated tools have been developed:

- **ProVerif** [1]: An automatic symbolic protocol verifier supporting an unbounded number of sessions, used to verify TLS 1.3 [17], Signal [16], and WireGuard.
- **Tamarin** [18]: A state-of-the-art protocol verifier that supports equational properties and inductive proofs, used for analyzing 5G authentication [19].

- **CryptoVerif** [2]: A computationally-sound protocol verifier that provides proofs in the computational model rather than the symbolic model.

Our work builds on these foundations, applying established verification techniques to the novel domain of local multi-agent coordination.

3.2 Capability-Based Authorization

Capability-based security dates back to Dennis and Van Horn’s seminal 1966 paper on programming semantics for multiprogrammed computations [7]. Recent work by Birgisson et al. [4] introduced Macaroons, which use chained HMACs for decentralized delegation with contextual caveats. The Anchor Protocol’s delegation chains (Section 5.3) are inspired by Macaroons but adapted for JWT-based identity in local development environments.

3.3 JWT Security

JSON Web Tokens have been the subject of extensive security analysis. Algorithm confusion attacks were first systematically studied by McLean [10], with subsequent vulnerabilities documented in CVE-2015-9235 [11] (HS256/RS256 confusion), CVE-2018-1000531 [12] (“none” algorithm), and CVE-2023-48238 [13] (generic algorithm confusion). Our protocol implements the defense recommended by the IETF JWT Best Current Practices [14]: strict algorithm whitelisting and key separation.

3.4 Credential Formats in Agentic Commerce

Since this protocol was first specified, the agentic-payments industry has converged on a concrete credential dialect: the Agent Payments Protocol [28], adopted by the Universal Commerce Protocol [27] (Google/Shopify, 2026), carries its authorization mandates as W3C Verifiable Credentials in SD-JWT form — SD-JWT+kb for principal proof, JWS detached-content signatures (RFC 7515 Appendix F) over JCS-canonicalized payloads (RFC 8785) for counterparty authorization, with ES256 as the recommended algorithm and keys published as JWK sets in a `.well-known` profile. The reference implementation migrates the Harbor Card envelope to this profile (hv: 3, ADR-0094) so that external verifiers need no bespoke SDK. The migration is deliberately envelope-only: the delegation-chain semantics, the attenuation invariant of Theorem 4.1, and the ProVerif obligations are independent of the serialization, though the key-binding construction of SD-JWT+kb adds a binding obligation the model must re-discharge. The JWT hardening above (strict algorithm whitelisting, key separation) transfers unchanged — SD-JWT is a JWT profile, not a departure from one.

4 Formal Verification Strategy

Following the industry standard set by AWS (s2n-tls [20]) and Microsoft (Project Everest [22]), we decoupled the verification of the *protocol design* from the verification of the *implementation*. The full stack has three layers (Figure 7): symbolic protocol analysis at the top, bounded model checking on the Rust source in the middle, and runtime conformance tests against the deployed binary at the bottom. Each layer’s obligations flow downward; each layer’s evidence flows back up. The stack is sound only if every bridge — symbolic-to-source and source-to-deployed — holds.

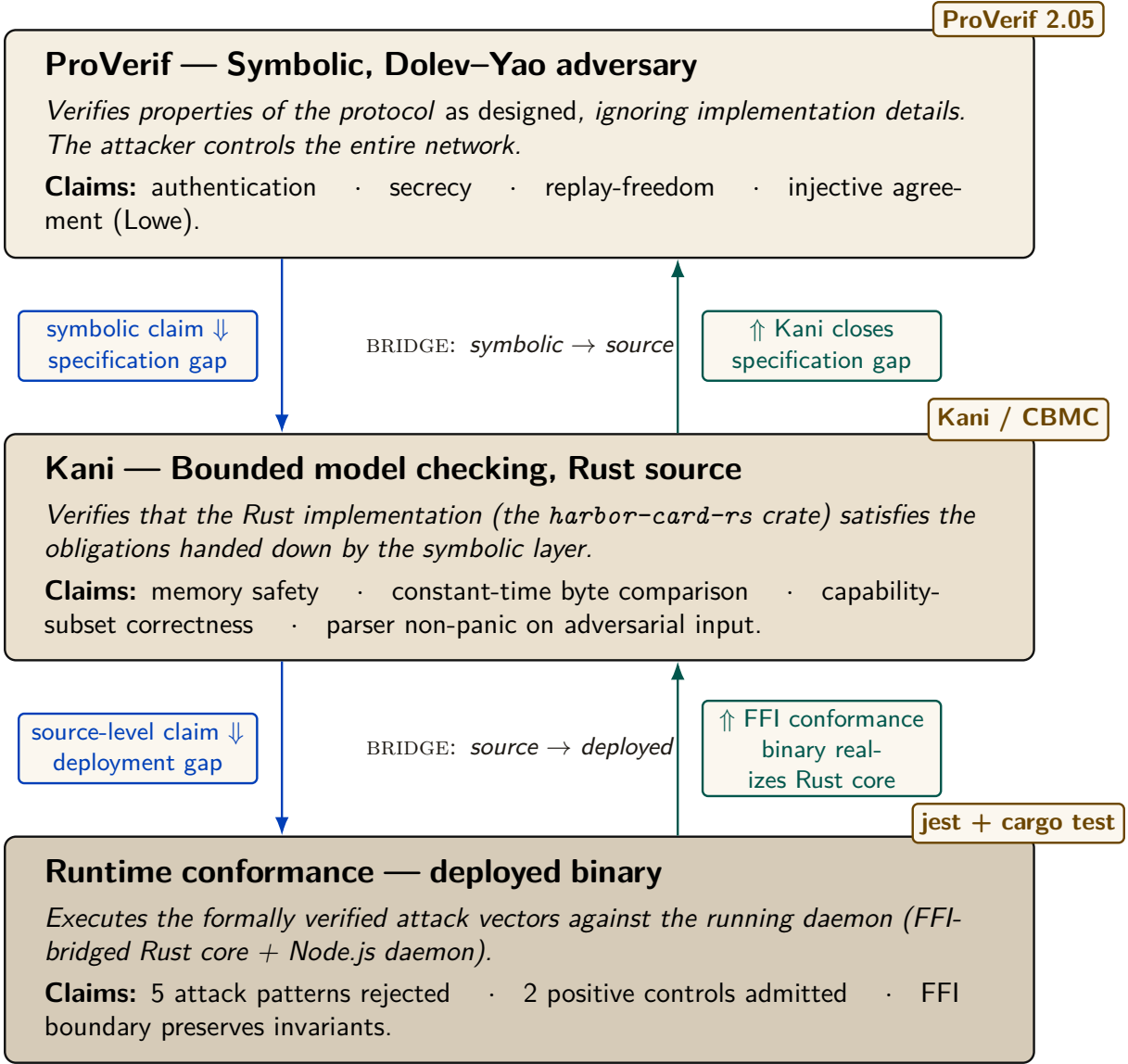


Figure 7: The three-layer verification stack. ProVerif (top) discharges design-level obligations against a Dolev–Yao adversary. Kani (middle) closes the *specification gap*: the same properties, restated as Rust source-level claims on the `harbor-card-rs` crate, are discharged by bounded model checking. Runtime conformance tests (bottom) close the *deployment gap*: the FFI-bridged binary is exercised with the formally verified attack vectors. Cinnabar arrows show obligations flowing down (what each lower layer must preserve); teal arrows show evidence flowing back up (what each lower layer actually delivers). The stack is sound only if every bridge holds.

4.1 Symbolic Analysis with ProVerif

To prove the protocol’s logical soundness against a Dolev–Yao adversary [6] (an attacker who controls the entire network), we modeled the Anchor Protocol in **ProVerif** [1]. ProVerif represents protocols as Horn clauses and uses resolution to prove security properties for an unbounded number of sessions.

Theorem 4.1 (Injective Agreement). For the multi-hop delegation chain, ProVerif confirmed that:

$$\begin{aligned} \text{Accepted}(b, h, \text{cap}) \implies & (\text{IssuedRoot}(a, h, \text{cap}) \wedge \text{Delegated}(a, b, \text{cap})) \\ & \vee \text{IssuedRoot}(b, h, \text{cap}) \end{aligned} \tag{1}$$

This *injective agreement* property [5] guarantees that capabilities cannot be escalated and trust is perfectly transitive.

All three protocol phases were independently verified in ProVerif 2.05. Table 1 summarizes the results; full models are reproduced in Appendices B, C, and D.

Query	Phase	Model	Result
$\neg \text{attacker}(\text{master_key})$	v1 (HS256)	v1_refined.pv	TRUE
Accepted $\implies$ Issued	v1 (HS256)	v1_refined.pv	TRUE
$\neg \text{attacker}(\text{harbor_sk})$	v2 (Ed25519)	v2_asymmetric.pv	TRUE
Accepted $\implies$ Issued	v2 (Ed25519)	v2_asymmetric.pv	TRUE
Accepted(b) $\implies$ IssuedRoot(a) $\wedge$ Delegated(a, b)	v3 (Delegation)	v3_delegation.pv	TRUE

Table 1: ProVerif verification results across all three protocol phases. All queries verified for an unbounded number of sessions under the Dolev-Yao adversary model.

The following is the verbatim ProVerif 2.05 output for the delegation chain query (Phase 3), the most complex property:

```

1 RESULT event(Accepted(b_2,harbor_id[],cap_1))
2 ==> (event(IssuedRoot(a_3,harbor_id[],cap_1))
3      && event(Delegated(a_3,b_2,cap_1)))
4      || event(IssuedRoot(b_2,harbor_id[],cap_1))
5 is true.

```

Listing 1: ProVerif 2.05 Terminal Output — Phase 3 Delegation Chain

4.2 Runtime Enforcement: The Arbiter

Formal proofs guarantee protocol *design* correctness. To enforce these guarantees at *runtime*, Port Daddy deploys the **Arbiter**—an ambient security agent that subscribes to the daemon’s activity log and checks every state transition against six formally derived invariants:

1. **PID_SQUATTING**: Detects when a process claims a port previously held by a different PID (the “Ghost in the Harbor”).
2. **CAP_ESCALATION**: Verifies capability subsets via the deployed Rust FFI enforcer.
3. **NOTE_MONOTONICITY**: Ensures the note sequence for active sessions never shrinks (from the TLA^+ *NoteMonotonicity* invariant).
4. **ESCROW_POSITIVE**: Validates that active sessions have positive escrow (when Float Plans are active).
5. **LOCK_OWNER_VALID**: Ensures locks are only held by registered agents with active sessions.
6. **HEARTBEAT_FRESHNESS**: Flags stale heartbeats as early warning for agent death.

Violations are logged to the activity trail and, in strict mode, trigger the “man overboard” salvage protocol. The Arbiter’s API (`GET /arbiter/status`, `GET /arbiter/violations`) provides real-time visibility into the security state of the commons.

4.3 Implementation Verification

A secure protocol is useless if the implementation contains buffer overflows or timing leaks. We extracted the critical JWT parsing and cryptographic comparison logic into a verified core.

Property 4.1 (Memory Safety). Using the **Kani** Rust Verifier [21], we performed bounded model checking to prove that our parsing logic never panics or accesses out-of-bounds memory, regardless of the input payload.

Property 4.2 (Side-Channel Resistance). To prevent timing attacks where an adversary guesses signatures byte-by-byte [23], we implemented a custom constant-time byte comparator. Kani formally verified that this comparator is branch-free regarding the contents of the secret arrays.

5 Verification Algorithms

In this section, we present the verification algorithms in pseudocode format, following the notation used in protocol verification literature [1, 5]. These algorithms correspond to the formal ProVerif models in Appendix B.

5.1 Phase 1: Symmetric HS256 with Algorithm Pinning

Algorithm 2 presents the secure verification procedure that is immune to algorithm confusion attacks (CVE-2015-9235 [11]).

```

1 Require: Pinned Algorithm: hs256
2 Require: Master Key: k_master (secret)
3
4 function Daemon.IssueToken(agent_id, harbor_id, capability):
5     jti <- GenerateNonce()
6     msg <- Encode(agent_id, harbor_id, jti, capability)
7     sigma <- HMAC-SHA256(msg, k_master)
8     emit Issued(agent_id, harbor_id, jti, capability)
9     return (hs256, msg, sigma)
10
11 function Harbor.VerifyToken(tok, expected_harbor):
12     (alg_header, msg, sigma) <- tok
13     // CRITICAL: Ignore alg_header, pin to HS256
14     if HMAC-SHA256(msg, k_master) != sigma:
15         return _|_ // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return _|_ // Wrong harbor
19     emit Accepted(a, h, j, cap)
20     return (a, h, j, cap)

```

Listing 2: Algorithm 1: Secure Harbor Verification (HS256 with Algorithm Pinning)

Security Property: The algorithm ignores the user-provided alg_{header} (line 15), preventing algorithm confusion attacks where an attacker sets $alg = \text{“none”}$ or attempts HS256/RS256 confusion [10].

5.2 Phase 2: Asymmetric Ed25519

Algorithm 3 presents the Ed25519-based verification. This phase removes the need for shared HMAC secrets between distributed Harbors.

```

1 Require: Harbor Keypair: (sk_harbor, pk_harbor)
2 Require: Key Distribution Channel: c_key (private)
3
4 function Daemon.IssueToken(agent_id, harbor_id, capability):
5     sk_h <- Receive(c_key) // Get harbor's secret key
6     jti <- GenerateNonce()
7     msg <- Encode(agent_id, harbor_id, jti, capability)
8     sigma <- Ed25519.Sign(msg, sk_h)
9     emit Issued(agent_id, harbor_id, jti, capability)
10    return (msg, sigma)
11
12 function Harbor.VerifyToken(tok, pk_h, expected_harbor):
13     (msg, sigma) <- tok
14     if not Ed25519.Verify(msg, sigma, pk_h):
15         return |_ // Invalid signature
16     (a, h, j, cap) <- Decode(msg)
17     if h != expected_harbor:
18         return |_ // Wrong harbor
19     emit Accepted(a, h, j, cap)
20    return (a, h, j, cap)

```

Listing 3: Algorithm 2: Asymmetric Harbor Verification (Ed25519)

Security Property: The secrecy of sk_{harbor} and the unforgeability of Ed25519 [8] ensure that only the legitimate Daemon can issue valid Harbor Cards.

5.3 Phase 3: Multi-hop Delegation

Algorithm 4 presents the delegation chain verification, inspired by Macaroons [4].

```

1 Require: Daemon Public Key: pk_daemon
2 Require: Subset Relation: subseteq on capabilities
3
4 function Agent.Delegate(token_parent, sk_parent, agent_child, cap_sub):
5     (msg_parent, sigma_parent) <- token_parent
6     (_, agent_parent, harbor, cap_parent) <- Decode(msg_parent)
7     if cap_sub not_subseteq cap_parent:
8         return |_ // Cannot escalate capabilities
9     msg_delegate <- Encode_delegation(msg_parent, sigma_parent,
10         agent_child, cap_sub)
11     sigma_delegate <- Ed25519.Sign(msg_delegate, sk_parent)
12     emit Delegated(agent_parent, agent_child, cap_sub)
13     return (msg_delegate, sigma_delegate)
14
15 function Harbor.VerifyChain(chain, pk_root):
16     tok_root <- chain[0]
17     (msg_0, sigma_0) <- tok_root
18     if not Ed25519.Verify(msg_0, sigma_0, pk_root):
19         return |_ // Invalid root signature
20     cap_root <- ExtractCaps(msg_0)
21     pk_current <- pk_root
22     for i <- 1 to |chain| - 1:
23         (msg_i, sigma_i) <- chain[i]
24         if not Ed25519.Verify(msg_i, sigma_i, pk_current):
25             return |_ // Invalid delegation signature
26         cap_i <- ExtractCaps(msg_i)
27         if cap_i not_subseteq cap_root:

```

```

27         return |_| // Capability escalation detected
28         pk_current <- ExtractPubkey(msg_i)
29         emit Accepted(agent_final, harbor, cap_final)
30     return T

```

Listing 4: Algorithm 3: Delegation Chain Verification

Security Property: The subset check on line 7 and line 22 ensures *capability attenuation*—delegated capabilities can only be restricted, never expanded. This is the mechanized content of Theorem D.1 (§D.1): under an enriched model with a real capability order and an active insider escalation adversary, ProVerif proves no escalating delegation is ever accepted, and a negative control confirms the guard is load-bearing. Combined with Theorem 4.1, this guarantees transitive trust.

6 Implementation: Deployed Rust Core

The verified cryptographic core is not a reference implementation—it is the **deployed production code**. The open-source Rust crate `harbor-card-rs` is compiled to a C dynamic library (`cdylib`) and called from the Node.js daemon via FFI through the daemon’s runtime-enforcement (Arbiter) module.

This architecture eliminates the gap between “verified code” and “running code” that weakens many formal verification efforts. The same binary that Kani model-checks is the same binary that the daemon loads at runtime.

6.1 Core Verification Logic

```

1 pub fn verify(&self, token: &str, now_ts: i64)
2     -> Result<HarborCardClaims, HarborError> {
3     let parts: Vec<&str> = token.split('.').collect();
4     if parts.len() != 3 {
5         return Err(HarborError::Malformed);
6     }
7     let msg = format!("{}", parts[0], parts[1]);
8     let mut sig_bytes = Self::internal_decode_b64(parts[2])?;
9     let sig_result = self.internal_verify_sig(
10         msg.as_bytes(), &sig_bytes);
11     sig_bytes.zeroize(); // Scrub signature from memory
12     sig_result?;
13     let mut payload_bytes = Self::internal_decode_b64(parts[1])?;
14     let claims: HarborCardClaims =
15         serde_json::from_slice(&payload_bytes)?;
16     payload_bytes.zeroize(); // Scrub payload from memory
17     if claims.exp < now_ts {
18         return Err(HarborError::Expired);
19     }
20     Ok(claims)
21 }
22
23 pub fn constant_time_compare(a: &[u8], b: &[u8]) -> bool {
24     if a.len() != b.len() { return false; }
25     let mut result = 0u8;
26     for i in 0..a.len() { result |= a[i] ^ b[i]; }
27     result == 0
28 }

```

Listing 5: Deployed Rust Implementation — Harbor Card Verifier (abridged from the `harbor-card-rs` crate)

Key implementation details: **zeroize** scrubs cryptographic material from memory after use, preventing heap inspection attacks. The constant-time comparator uses XOR accumulation rather than early-return, ensuring execution time is independent of input content.

6.2 FFI Bridge to the Node.js Daemon

The Rust crate exports two C-ABI functions consumed by the TypeScript daemon:

```

1 #[no_mangle]
2 pub extern "C" fn harbor_constant_time_compare(
3     a: *const u8, a_len: usize,
4     b: *const u8, b_len: usize
5 ) -> bool { /* ... */ }
6
7 #[no_mangle]
8 pub extern "C" fn harbor_verify_caps_subset_json(
9     root_json: *const c_char, root_len: usize,
10    sub_json: *const c_char, sub_len: usize
11 ) -> bool { /* ... */ }

```

Listing 6: FFI Exports — Called from the daemon’s Arbiter module

The daemon’s arbiter module binds these at startup:

```

1 constantTimeCompare: lib.func(
2     'bool harbor_constant_time_compare(
3         const uint8_t *a, size_t a_len,
4         const uint8_t *b, size_t b_len)')',
5 verifyCapsSubset: lib.func(
6     'bool harbor_verify_caps_subset_json(
7         const char *root_json, size_t root_len,
8         const char *sub_json, size_t sub_len)')'

```

Listing 7: TypeScript FFI Binding (daemon-side Arbiter)

6.3 Kani Verification Harnesses

Three bounded model checking proofs are defined in the same source file under `#[cfg(kani)]`:

```

1 #[cfg(kani)]
2 #[kani::proof]
3 #[kani::stub(HarborCardVerifier::internal_pk_from_bytes,
4     stubs::pk_from_bytes_stub)]
5 #[kani::stub(HarborCardVerifier::internal_decode_b64,
6     stubs::decode_b64_stub)]
7 #[kani::stub(HarborCardVerifier::internal_verify_sig,
8     stubs::verify_sig_stub)]
9 #[kani::unwind(10)]
10 fn proof_verify_logic_only() {
11     let pk_bytes: [u8; 32] = kani::any();
12     if let Ok(verifier) = HarborCardVerifier::new(pk_bytes) {
13         let token_bytes: [u8; 32] = kani::any();
14         if let Ok(token_str) =
15             std::str::from_utf8(&token_bytes) {
16             kani::assume(token_str.contains('.')');
17             let _ = verifier.verify(token_str, 0);
18         }
19     }
20 }

```



```

21
22 #[cfg(kani)]
23 #[kani::proof]
24 fn proof_constant_time_behavior() {
25     let a: [u8; 16] = kani::any();
26     let b: [u8; 16] = kani::any();
27     let _ = HarborCardVerifier::constant_time_compare(&a, &b);
28 }
29
30 #[cfg(kani)]
31 #[kani::proof]
32 fn proof_capability_attenuation() {
33     let root = vec!["read".to_string(), "write".to_string()];
34     let mut sub = vec!["read".to_string()];
35     assert!(HarborCardVerifier::verify_capability_subset(
36         &root, &sub));
37     sub.push("admin".to_string());
38     assert!(HarborCardVerifier::verify_capability_subset(
39         &root, &sub));
40 }

```

Listing 8: Kani Proof Harnesses (Deployed Source)

The stubbing strategy deserves explanation: `proof_verify_logic_only` stubs out Ed25519 signature verification and base64 decoding (which involve curve arithmetic that Kani cannot symbolically execute within practical bounds) and exhaustively explores the *parsing and control flow logic*—the split-on-dots, payload extraction, expiry comparison, and error handling paths. This verifies that the logic surrounding the cryptographic primitives never panics, never accesses out-of-bounds memory, and handles all malformed inputs gracefully, regardless of what the cryptographic layer returns.

The build output at `target/kani/aarch64-apple-darwin/debug` contains the CBMC intermediate representations used by Kani’s bounded model checker.

7 Security Analysis

7.1 Threat Model

We assume a **Dolev-Yao adversary** [6] who:

- Controls the entire network
- Can intercept, modify, and inject messages
- Cannot break cryptographic primitives (cryptography is perfect)
- May corrupt legitimate agents (but not all simultaneously)

This is the standard threat model for symbolic protocol analysis [1].

7.2 Security Properties

Property	Verified	Tool	Deployed	Reference
Master Key Secrecy	Yes	ProVerif	—	[1]
Algorithm Confusion Immunity	Yes	ProVerif	—	[10]
Authentication (Injective Agreement)	Yes	ProVerif	—	[5]
Delegation Integrity	Yes	ProVerif	—	[4]
Memory Safety (parsing)	Yes	Kani	Yes (FFI)	[21]
Constant-Time Comparison	Yes	Kani	Yes (FFI)	[23]
Capability Attenuation	Yes	Kani	Yes (FFI)	[4]

Table 2: Verified Security Properties. “Deployed” indicates the verified Rust code is called by the production daemon via FFI from the Arbiter module into the `harbor-card-rs` crate.

7.3 Limitations

1. **Delegation-chain depth.** The symbolic proof of delegation soundness (Table 3, “`chain-replay.pv`”) is machine-checked at the chain depth realized in deployment (depth 3); it is *not* a proof for unbounded chain length. ProVerif’s unbounded-session guarantee quantifies over the number of protocol *sessions*, not over the number of hops in a single delegation chain, which is a fixed structural bound in the modeled process. Extending the proof to unbounded depth (via recursive replication of the delegation process) is future work. The current depth bound is a *spawn-time* policy check (`assessDelegation` in `lib/tube-spawner-router.ts`, `HARD_MAX_DELEGATION_DEPTH=8`, default 4) applied when a spawn request arrives; it is *not* a verification-time invariant. The cryptographic chain verifier (`lib/delegation-chain.ts`) accepts arbitrary chain depth, and the attenuation monitor (`lib/cap-attenuation-monitor.ts`) enforces only per-hop capability monotonicity—not total depth. Machine-enforced depth at verification time is a planned hardening item.
2. **PID Binding:** The “Ghost in the Harbor” scenario requires binding tokens to the requesting PID, which is handled at the OS level.
3. **Local Transport Security:** On multi-user systems, local unix sockets or loopback TLS should be used to prevent local bearer token sniffing.

8 Conclusion

The Anchor Protocol elevates local multi-agent development from a state of “hope-based security” to “math-based security.” By combining:

- Symbolic protocol proofs (ProVerif [1])
- Memory-safe implementation (Rust/Kani [21])
- Capability-based delegation (inspired by Macaroons [4])

Port Daddy provides a formally verified control plane capable of safely orchestrating the next generation of autonomous AI swarms.

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A Formal Verification Status

Artifact availability. Every artifact named in Table 3 is bundled at <https://portdaddy.dev/whitepaper/artifacts/>. The verified Rust core is the open-source `harbor-card-rs` crate; runtime components are deployed inside the Port Daddy daemon binary. The companion paper, *The Bonded Commons*, carries the cross-paper status table; this appendix lists only the items that pertain specifically to the Anchor Protocol.

Table 3: Anchor Protocol verification status. **closed**: model verified *and* runtime conformance test passes. **partial**: design verified, runtime empirically tested rather than proved. **open**: known unmechanized.

Claim	Method	Result	Artifact
Authentication + injective agreement, all three phases (§5)	ProVerif 2.05	closed all three phases	harbor_card_v{1,2,3}*.pv
Algorithm-confusion immunity	ProVerif pinned vs. naive	closed pinned TRUE; counter-trace witnessed	algconfusion.pv
Delegation chain replay (§5.3)	ProVerif at depth 3	closed chain authenticity TRUE; 17-case runtime conformance	chain-replay.pv
Cuckoo-filter pollution resistance (§2.4)	5-invariant property test	closed FP rate within $2B/2^f$ at load 0.95	cuckoo property tests
GitHub-App webhook origin authentication	ProVerif (Dolev-Yao relay)	closed origin auth TRUE; daemon re-verifies GitHub HMAC (runtime conformance)	github_ingress_origin_auth.pv
Multi-tenant relay namespace isolation	ProVerif + runtime	closed member of A cannot publish into B ; HARBOR_MISMATCH conformance	github_ingress_tenant_isolation
Cuckoo-filter revocation soundness	inspection + empirical	partial gate strictly narrows acceptance; gossip bound reproduced from [26]	runtime test
Constant-time signature comparison	audited library (<code>subtle::ConstantTimeEq</code>)	partial no side-channel proof; library trust documented	—
Relay payload secrecy via HPKE (daemon key pinned)	ProVerif (design)	partial secrecy TRUE under pinned key; seal not yet implemented	github_ingress_hpke_secrecy.pv
Webhook replay-freedom + in-order delivery	TLA+ / TLC (exhaustive at bound)	partial design model-checked; runtime via ordered-chain ingest	WebhookDelivery.tla
Card revocation finality under backup/restore (ADR-0018 Attack 2)	TLA+ / TLC	partial roll-back attack + revocation-epoch mitigation both model-checked	CardRevocation.tla
Formal gossip-convergence bound	—	open leaning on standard anti-entropy analysis [26]	—

Limitations. The cuckoo-filter pollution closure handles the local-filter side; cross-peer gossip propagation still leans on the standard anti-entropy analysis rather than a mechanized proof. The side-channel resistance of signature comparison is inherited from an audited library rather than re-proved. These deferrals are documented as known open problems rather than as defects in the present claims.

GitHub-App relay ingress. The webhook-dispatch surface (GitHub → untrusted relay → daemon) is modeled with the relay as the Dolev-Yao attacker, in three relay-agnostic properties, each paired with a negative control (`*_vuln.pv`) that exhibits the attack once the mitigation is removed—so a *true* result is non-vacuous. (i) *Origin authentication*: the daemon dispatches a fleet event only if GitHub signed it, even when the forwarder’s transport credential is attacker-held; the control omits the daemon-side HMAC re-verification and the property fails (forged dispatch). This is *non-injective* agreement (origin), not replay-freedom: a captured genuine (*payload*, *mac*) may be re-delivered. Replay-freedom is an ordering-sensitive, mutable-state property outside ProVerif’s reach, so it is discharged separately in TLA+ (`proofs/relay/WebhookDelivery.tla`): a daemon consuming the *ordered* per-publisher chain (dispatch iff *seq* = *tip* + 1) is model-checked at-most-once *and* gap-free against an adversarial relay that reorders, duplicates, and redelivers, with the unguarded daemon as the model-checked counterexample (TLC exhibits a double-dispatch). The obligation it names: the daemon ingests webhooks over the relay’s ordered `from_seq` subscribe path, not the raw forward. (ii) *Payload secrecy*: under HPKE to the daemon’s *pinned* X25519 key the relay never learns plaintext; the control sources the key from the relay and the secrecy property collapses to a key-substitution MITM—hence out-of-band key pinning is a requirement, not a recommendation. The seal is specified but not yet implemented, so this row is **partial**. (iii) *Namespace isolation*: a member of tenant *A* cannot publish into tenant *B* because harbor binding is checked against authoritative membership; the control trusts the card’s self-declared issuer and crosses the tenant boundary. Honesty rider: HMAC-SHA256 gives origin authentication to a shared-key holder (EUF-CMA), not third-party non-repudiation; HPKE base mode is assumed IND-CCA2.

We err toward listing fewer items as closed rather than more.

B Full ProVerif Models

For completeness, we include the full ProVerif models used for verification. These correspond to the algorithms in Section 5.

B.1 Phase 1: Symmetric HS256

```

1 (* Port Daddy: Harbor Card v1 (HS256) Formal Model *)
2 type id. type key. type jti. type capability. type alg_type.
3 const hs256_alg: alg_type. const none_alg: alg_type.
4 free c: channel.
5
6 fun hs256(bitstring, key): bitstring.
7   reduc forall m: bitstring, k: key;
8     check_hs256(m, k, hs256(m, k)) = true.
9
10  reduc forall m: bitstring, k: key;
11    verify_generic(m, hs256_alg, hs256(m, k), k) = true;
12    forall m: bitstring, k: key;
13    verify_generic(m, none_alg, m, k) = true.
```

```

14
15 fun encode_token(id, id, jti, capability): bitstring [data].
16
17 event Issued(id, id, jti, capability).
18 event Accepted(id, id, jti, capability).
19
20 free master_key: key [private].
21 query attacker(master_key).
22
23 query a: id, h: id, j: jti, cap: capability;
24   event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).
25
26 let Daemon(k: key, a_id: id, h_id: id, j: jti, cap: capability) =
27   let msg = encode_token(a_id, h_id, j, cap) in
28   let signature = hs256(msg, k) in
29   event Issued(a_id, h_id, j, cap);
30   out(c, (hs256_alg, msg, signature)).
31
32 let SecureHarbor(k: key, expected_h: id) =
33   in(c, (alg_header: alg_type, msg: bitstring, signature: bitstring))
34   ;
35   if check_hs256(msg, k, signature) = true then
36     let encode_token(a: id, h: id, j: jti, cap: capability) = msg in
37     if h = expected_h then
38       event Accepted(a, h, j, cap).
39
40 process
41   new agent_id: id; new harbor_id: id;
42   new token_jti: jti; new secret_cap: capability;
43   (!Daemon(master_key, agent_id, harbor_id, token_jti, secret_cap) |
44     !SecureHarbor(master_key, harbor_id))

```

Listing 9: harbor_card_v1_refined.pv

C Phase 2: Asymmetric Ed25519

```

1 (* Port Daddy: Harbor Card v2 (Ed25519 / EdDSA) Formal Model *)
2 type id. type skey. type pkey. type jti. type capability.
3
4 free c: channel.
5 free k_dist: channel [private]. (* Trusted key distribution *)
6
7 (** Digital Signatures (Ed25519) **)
8 fun pk(skey): pkey.
9 fun sign(bitstring, skey): bitstring.
10 reduc forall m: bitstring, k: skey;
11   check_sign(m, pk(k), sign(m, k)) = true.
12
13 fun encode_token(id, id, jti, capability): bitstring [data].
14
15 (** Events **)
16 event Issued(id, id, jti, capability).
17 event Accepted(id, id, jti, capability).
18
19 (** Queries **)
20 free secret_key: skey [private].
21 query attacker(secret_key).

```

```

22
23 query a: id, h: id, j: jti, cap: capability;
24   event(Accepted(a, h, j, cap)) ==> event(Issued(a, h, j, cap)).
25
26 (** Processes **)
27 let Daemon(a_id: id, h_id: id, j: jti, cap: capability) =
28   in(k_dist, sk_h: skey);
29   let msg = encode_token(a_id, h_id, j, cap) in
30   let signature = sign(msg, sk_h) in
31   event Issued(a_id, h_id, j, cap);
32   out(c, (msg, signature)).
33
34 let Harbor(pk_h: pkey, expected_h: id) =
35   in(c, (msg: bitstring, signature: bitstring));
36   if check_sign(msg, pk_h, signature) = true then
37     let encode_token(a: id, h: id, j_1: jti, cap_1: capability)
38       = msg in
39     if h = expected_h then
40       event Accepted(a, h, j_1, cap_1).
41
42 (** Main Process **)
43 process
44   new agent_id: id; new harbor_id: id;
45   new token_jti: jti; new secret_cap: capability;
46   let harbor_pk = pk(secret_key) in
47   out(c, harbor_pk);
48   (
49     (!Daemon(agent_id, harbor_id, token_jti, secret_cap)) |
50     (!Harbor(harbor_pk, harbor_id)) |
51     (!out(k_dist, secret_key))
52   )

```

Listing 10: harbor_card_v2_asymmetric.pv — Verified in ProVerif 2.05

D Phase 3: Multi-hop Delegation

```

1 (* Port Daddy: Harbor Card v3 (Delegation Chains) Formal Model *)
2 type id. type skey. type pkey. type capability.
3
4 free c: channel.
5
6 (** Cryptographic Functions **)
7 fun pk(skey): pkey.
8 fun sign(bitstring, skey): bitstring.
9 reduc forall m: bitstring, k: skey;
10   check_sign(m, pk(k), sign(m, k)) = true.
11
12 reduc forall c1: capability; is_subset(c1, c1) = true.
13
14 fun base_token(id, id, id, capability): bitstring [data].
15 fun delegate_token(bitstring, id, capability): bitstring [data].
16
17 (** Events **)
18 event IssuedRoot(id, id, capability).
19 event Delegated(id, id, capability).
20 event Accepted(id, id, capability).
21

```



```

22 (** Queries **)
23 free harbor_id: id.
24 query b: id, cap: capability, a: id;
25   event(Accepted(b, harbor_id, cap)) ==>
26     (event(IssuedRoot(a, harbor_id, cap))
27      && event(Delegated(a, b, cap)))
28     || event(IssuedRoot(b, harbor_id, cap)).
29
30 (** Processes **)
31 free daemon_id: id.
32 free daemon_sk: skey [private].
33
34 let Daemon(a: id, cap: capability) =
35   let msg = base_token(daemon_id, a, harbor_id, cap) in
36   let s = sign(msg, daemon_sk) in
37   event IssuedRoot(a, harbor_id, cap);
38   out(c, (msg, s)).
39
40 let AgentA(a: id, a_sk: skey, b: id, sub_cap: capability) =
41   in(c, (msg: bitstring, s: bitstring));
42   let base_token(=daemon_id, =a, =harbor_id,
43     root_cap: capability) = msg in
44   if check_sign(msg, pk(daemon_sk), s) = true then
45     if is_subset(sub_cap, root_cap) = true then
46       let d_msg = delegate_token((msg, s), b, sub_cap) in
47       let d_s = sign(d_msg, a_sk) in
48       event Delegated(a, b, sub_cap);
49       out(c, (d_msg, d_s)).
50
51 let Harbor(a_pk: pkey) =
52   in(c, (d_msg: bitstring, d_s: bitstring));
53   let delegate_token((base_msg: bitstring, base_s: bitstring),
54     b: id, sub_cap: capability) = d_msg in
55   let base_token(=daemon_id, a: id, =harbor_id,
56     root_cap: capability) = base_msg in
57   if check_sign(base_msg, pk(daemon_sk), base_s) = true then
58     if check_sign(d_msg, a_pk, d_s) = true then
59       if is_subset(sub_cap, root_cap) = true then
60         event Accepted(b, harbor_id, sub_cap).
61
62 (** Main Process **)
63 process
64   new sk_a: skey;
65   let pk_a = pk(sk_a) in
66   out(c, pk_a);
67   new id_a: id; new id_b: id; new cap_root: capability;
68   (
69     (!Daemon(id_a, cap_root)) |
70     (!AgentA(id_a, sk_a, id_b, cap_root)) |
71     (!Harbor(pk_a))
72   )

```

Listing 11: harbor_card_v3_delegation.pv — reflexive-order model (see §D.1 for the soundness caveat and the enriched v5 proof)

D.1 Soundness of the Attenuation Proof (v5)

The v3 model above is honest about cryptographic structure but *vacuous about attenuation*. Its capability order is reflexive only:

```
reduc forall c1: capability; is_subset(c1, c1) = true.
```

Capabilities are opaque atoms with no order, so a delegated capability can only ever *equal* its parent. Escalation—delegating more authority than you hold—is not blocked; it is *inexpressible*. The no-escalation query therefore holds for a reason that has nothing to do with the protocol: there is no larger capability to delegate. A reviewer flagged this directly: “*the model cannot witness escalation.*”

`harbor_card_v5_attenuation.pv` closes the obligation by enrichment, not by weakening the claim. It gives capabilities a real partial order (`cap_read` $\sqsubseteq$ `cap_write`, both public so the attacker can construct either), replaces the reflexive stub with the sound order (`is_subset(cap_read, cap_write)` derivable; `is_subset(cap_write, cap_read)` *not*), and adds an *insider escalation adversary*: an agent holding a valid `cap_read` root token that signs an over-broad `cap_write` delegation with no self-restraint. The verifier’s `is_subset` guard is the sole defense under test.

```

1 (* Sound order: reflexive + read <= write; write <= read NOT
   derivable *)
2 fun is_subset(capability, capability): bool
3   reduc
4     forall x: capability; is_subset(x, x) = true
5     otherwise is_subset(cap_read, cap_write) = true.
6
7 (* Insider adversary: holds a real root, signs an over-broad
   delegation *)
8 let MaliciousA(a: id, a_sk: skey, b: id) =
9   in(c, (msg: bitstring, s: bitstring));
10  let base_token(=daemon_id, =a, =harbor_id, root_cap: capability) =
11    msg in
12    if check_sign(msg, pk(daemon_sk), s) = true then
13      event EscalationAttempted(a, root_cap, cap_write);
14      let d_msg = delegate_token((msg, s), b, cap_write) in
15      out(c, (d_msg, sign(d_msg, a_sk))).
16
17 (* Q1 soundness: a write-card delegated from a read-root is NEVER
   accepted *)
18 query b: id; event(Accepted(b, harbor_id, cap_write, cap_read)) ==>
19   false.
20 (* Q2 non-vacuity: the escalation IS reachable (the proof is not
   vacuous) *)
21 query a: id, r: capability, s: capability;
22   event(EscalationAttempted(a, r, s)) ==> false.
```

Listing 12: `harbor_card_v5_attenuation.pv` (excerpt) — Verified in ProVerif 2.05

Theorem D.1 (Attenuation soundness, mechanized). In `harbor_card_v5_attenuation.pv`, ProVerif 2.05 reports `Accepted(b, harbor, cap_write, cap_read)` unreachable (**Q1 = true**): no escalating delegation is ever accepted. Simultaneously `EscalationAttempted` is reachable (**Q2 = false-with-trace**): the attack is expressible, so Q1 is a real defense, not the v3 vacuum. A negative control that deletes the verifier’s `is_subset` guard flips Q1 to false-with-trace—ProVerif exhibits the read→write escalation—confirming the guard is load-bearing.

Multi-hop scope (honest boundary). Theorem D.1 is single-hop (Daemon → A → verifier). Multi-hop delegation has a sharper obligation: a verifier that checks the *final* capability

against the *root* accepts a non-monotonic middle hop. `harbor_card_v6_multihop_attack.pv` mechanizes exactly this — $A:\text{write} \rightarrow B:\text{read} \rightarrow C:\text{write}$ is accepted (ProVerif: escalation reachable). The fix, `harbor_card_v7_multihop_fixed.pv`, verifies each hop against its *immediate parent* ($\text{is_subset}(\text{final}, \text{mid}) \wedge \text{is_subset}(\text{mid}, \text{root})$) and ProVerif proves the same chain rejected. The runtime delegation verifier — and the cross-harbor recompute $\text{att}_B \circ \text{att}_A$ — must be per-hop, never final-vs-root.